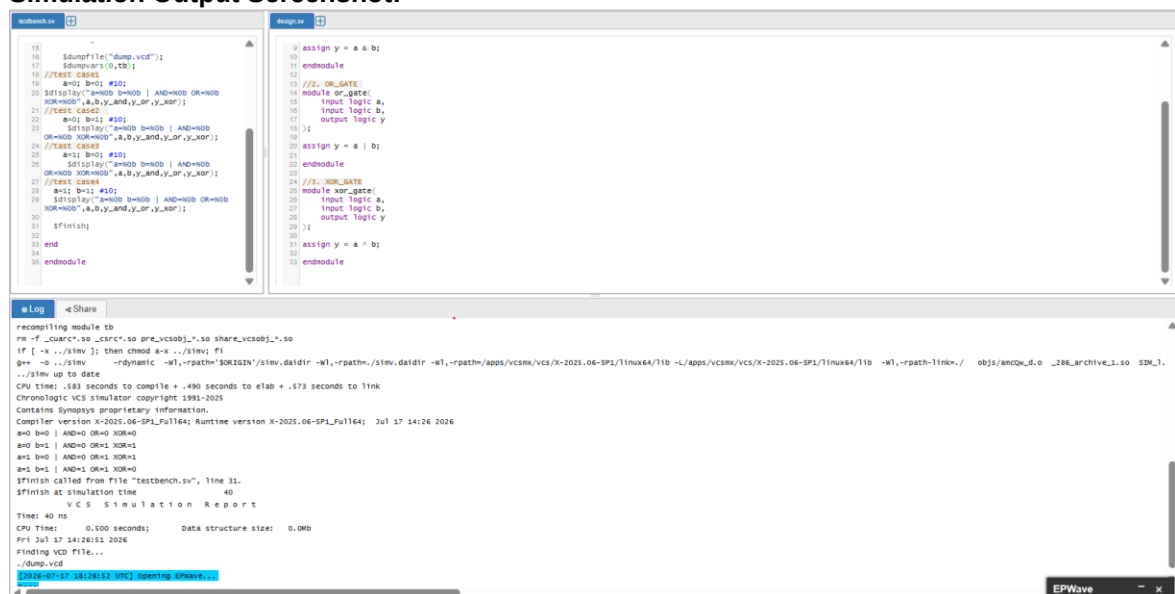
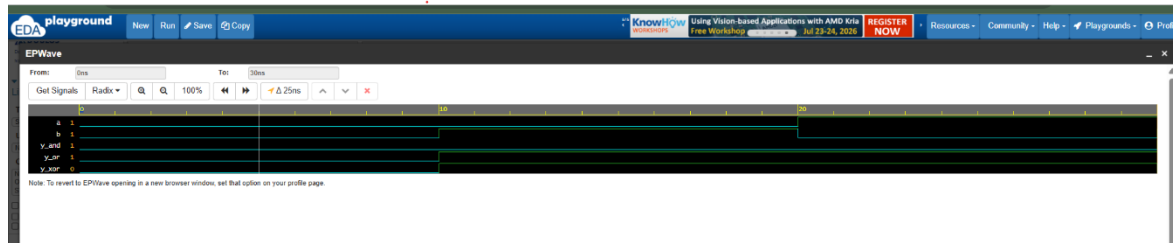




Waveform Screenshot:



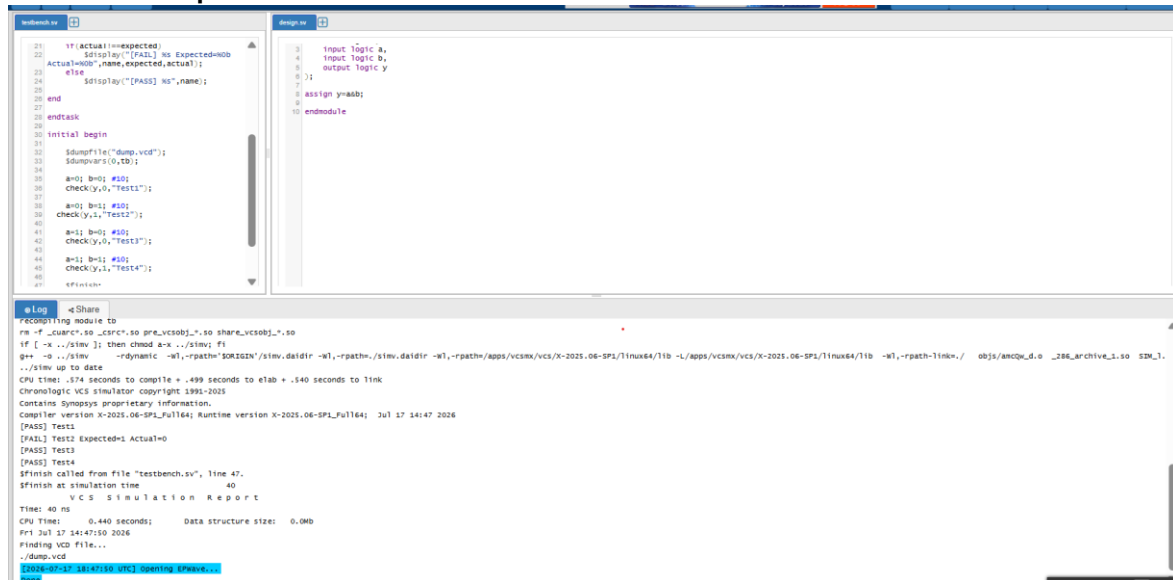
Explanation: The outputs matched the expected truth tables. AND produced 1 only for 11, OR produced 1 whenever at least one input was 1, and XOR produced 1 only when the inputs differed.

Task 2: Self-Checking Testbench

Objective: Implement a reusable `check()` task to automatically print PASS/FAIL for each test vector.

EDA Playground Link: <https://edaplayground.com/x/THmJ>

Simulation Output Screenshot:



Explanation: The testbench used the 4-state operator (!==) to compare the DUT output with the expected value. One incorrect expected value was intentionally introduced to demonstrate a FAIL message before restoring the correct implementation.

Conclusion

Section 5 introduced the EDA Playground workflow, module/testbench organization, simulation, waveform viewing, and self-checking verification. Both tasks were completed successfully after verifying the console output and waveforms.